

A second-predual counterexample for Gaussian random-field samples

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Source question

Korolev, Latz, and Schoenlieb ask in the Outlook of *Gaussian random fields on non-separable Banach spaces* whether the presence of a second predual can strengthen their bounded-pairing and weak-star-completion conclusions to actual membership of the sample in U .

Second preduals. First, we go back to Section 4.2. Here, we have encountered a situation where the space $U = \text{Lip}(\Omega)$ has a second predual, the little Lipschitz space. The same happens in Section 4.3 with $U = C^{**}(\Omega)$. A natural question is now, whether it is possible in this case to strengthen Theorems 3.2 and 3.3 and show that $\theta \in U$ with probability 1.

Random fields with jumps. In this work we have focused on Gaussian measures and the generalisation to other probability distributions is interesting. Of special interest are random samples with jumps, e.g. piecewise continuous or piecewise constant samples where the subdomains on which the sampled functions are continuous/constant are also random. Chada et al. [6], for instance, discuss Cauchy random fields. It would be highly interesting to study such non-Gaussian random fields on, e.g., spaces of functions of bounded variation.

Conditioning. Gaussian processes are of particular interest in Bayesian statistics and data science as it is possible in linear settings to determine conditional mean and covariance in closed form. Scovel and Owhadi [27] have studied the conditioning of Gaussian random fields on Hilbert spaces. A natural next goal is to generalise their theory to our setting of Banach spaces.

Figure 1: The source Outlook question, local build PDF page 14.

In the notation of the source paper, a positive nuclear covariance is written as

$$c = \sum_{i=1}^{\infty} \lambda_i \phi_i \otimes \phi_i, \quad \lambda_i \geq 0, \quad \sum_i \lambda_i < \infty,$$

with $\|\phi_i\|_U = 1$, and the associated formal Gaussian sample is

$$\theta = \sum_{i=1}^{\infty} \lambda_i^{1/2} \xi_i \phi_i,$$

where the ξ_i 's are independent standard Gaussian variables. The source paper proves that each fixed predual pairing is finite almost surely and that θ belongs to a weak-star completion. The question is whether, in second-predual cases, one can conclude $\theta \in U$ almost surely.

Proof intuition

The obstruction is that scalar testing by each fixed predual vector does not control a supremum over too many coordinates. In ℓ_{∞} , make the coordinates correspond to finite subsets $F \subset \mathbb{N}$. The i -th covariance direction is the indicator of the family of finite sets containing i . Then the coordinate indexed by F is the finite partial subseries $\sum_{i \in F} \xi_i / i$. Taking F to be the positive indices among $\{1, \dots, N\}$, these coordinates dominate a Bernoulli-thinned harmonic series, and hence become unbounded almost surely.

Counterexample

Theorem 1. *Let $U = \ell_\infty([\mathbb{N}]^{<\omega})$. Then $U = c_0([\mathbb{N}]^{<\omega})^{**}$, so U has the second predual $c_0([\mathbb{N}]^{<\omega})$. There are norm-one vectors $(u_i) \subset U$ and positive numbers $(\lambda_i) \in \ell_1$ such that the positive nuclear kernel*

$$c = \sum_{i=1}^{\infty} \lambda_i u_i \otimes u_i \in U \widehat{\otimes}_{\pi} U$$

has the source paper's scalar bounded-pairing property, but the corresponding Gaussian sample

$$\theta = \sum_{i=1}^{\infty} \lambda_i^{1/2} \xi_i u_i$$

does not belong to U almost surely.

Proof. Let $[\mathbb{N}]^{<\omega}$ be the countable set of all finite subsets of $\mathbb{N}$. For $i \in \mathbb{N}$, define $u_i \in \ell_\infty([\mathbb{N}]^{<\omega})$ by

$$u_i(F) = \begin{cases} 1, & i \in F, \\ 0, & i \notin F, \end{cases} \quad F \in [\mathbb{N}]^{<\omega}.$$

Then $\|u_i\|_\infty = 1$. Put $\lambda_i = i^{-2}$. Since $\sum_i \lambda_i < \infty$, the series

$$c = \sum_{i=1}^{\infty} i^{-2} u_i \otimes u_i$$

converges in the projective tensor product. It is symmetric and positive in the covariance sense, because for every $\eta \in \ell_1([\mathbb{N}]^{<\omega})$,

$$\langle \eta, C\eta \rangle = \sum_{i=1}^{\infty} i^{-2} \langle \eta, u_i \rangle^2 \geq 0.$$

Let

$$\theta(F) = \sum_{i \in F} \frac{\xi_i}{i}, \quad F \in [\mathbb{N}]^{<\omega}.$$

Each coordinate is a finite sum. Moreover, if $\eta \in \ell_1([\mathbb{N}]^{<\omega})$ and

$$b_i = \langle \eta, u_i \rangle = \sum_{F \ni i} \eta_F,$$

then $|b_i| \leq \|\eta\|_1$, so

$$\sum_{i=1}^{\infty} \frac{b_i^2}{i^2} \leq \|\eta\|_1^2 \sum_{i=1}^{\infty} \frac{1}{i^2} < \infty.$$

Hence the scalar Gaussian series

$$\sum_{i=1}^{\infty} \frac{\xi_i b_i}{i}$$

converges almost surely. This is exactly the bounded-pairing conclusion for each fixed predual vector η .

It remains to show that $\theta \notin \ell_\infty([\mathbb{N}]^{<\omega})$ almost surely. Let $B_i = \mathbf{1}_{\{\xi_i > 1\}}$. By the weighted strong law of large numbers, or equivalently by summation by parts from the ordinary strong law,

$$\frac{\sum_{i=1}^N B_i/i}{\sum_{i=1}^N 1/i} \longrightarrow \mathbb{P}(\xi_1 > 1) > 0 \quad \text{almost surely.}$$

Thus $\sum_i B_i/i = \infty$ almost surely.

For each N , set $F_N = \{1 \leq i \leq N : \xi_i > 1\}$. This is a finite subset of $\mathbb{N}$, hence $F_N \in [\mathbb{N}]^{<\omega}$, and

$$\|\theta\|_\infty \geq \theta(F_N) = \sum_{\substack{1 \leq i \leq N \\ \xi_i > 1}} \frac{\xi_i}{i} \geq \sum_{i=1}^N \frac{B_i}{i}.$$

The right-hand side is unbounded in N almost surely. Therefore $\theta \notin U$ almost surely. $\square$

Corollary 1 (Isomorphic transfer). *Let V be any Banach space linearly isomorphic to ℓ_∞ . Then there is a positive nuclear covariance kernel in $V \hat{\otimes}_\pi V$ for which all fixed predual pairings are finite almost surely, but the associated Gaussian sample is not in V almost surely.*

Proof. Let $T : \ell_\infty([\mathbb{N}]^{<\omega}) \rightarrow V$ be an isomorphism and use the vectors u_i from the theorem. Define

$$v_i = \frac{Tu_i}{\|Tu_i\|_V}, \quad \lambda_i = i^{-2}.$$

Then $\|v_i\|_V = 1$ and $\sum_i \lambda_i < \infty$, so $\sum_i \lambda_i v_i \otimes v_i$ is a positive nuclear kernel in $V \hat{\otimes}_\pi V$. If the corresponding Gaussian series belonged to V , applying T^{-1} would give a bounded element of $\ell_\infty([\mathbb{N}]^{<\omega})$ with coordinates

$$F \mapsto \sum_{i \in F} \frac{\xi_i}{i \|Tu_i\|_V}.$$

Since $\|Tu_i\|_V \leq \|T\|$, the same choice $F_N = \{1 \leq i \leq N : \xi_i > 1\}$ gives coordinates at least $\|T\|^{-1} \sum_{i \leq N} B_i/i$, which are unbounded almost surely. $\square$

Consequence for the source question

The source paper itself recalls that, under its Theorem 8.49 hypothesis, the spaces $\text{lip}(X^\alpha)$, $\text{AE}(X^\alpha)$, and $\text{Lip}(X^\alpha)$ are linearly homeomorphic to c_0 , ℓ_1 , and ℓ_∞ , respectively. The corollary therefore gives a counterexample in those Lipschitz-space instances: a second-predual structure does not by itself upgrade the source paper's weak-star conclusion to $\theta \in U$ almost surely.

Limitations

This packet does not rule out stronger positive theorems for special covariance kernels, for example the exponential kernels studied in the source paper, or for covariance operators satisfying the appropriate Banach-space Gaussian radonifying condition. It only refutes the broad strengthening based solely on the existence of a second predual.

Novelty and literature check

Before promotion, the cheap run indexes were searched for arXiv:2203.04650 and for the phrases “Gaussian random fields”, “non-separable Banach”, “second predual”, “nuclear covariance”, “ ℓ_∞ ”, and close variants. No duplicate packet or attempt was found.

External web searches were run for the exact source title, for “Gaussian random fields non-separable Banach second predual theta in U”, and for variants of “nuclear covariance Gaussian series ℓ_∞ not bounded”. The only exact source-title hit was the arXiv paper itself, and no later paper explicitly answering the Outlook question was found in this bounded search. The construction is elementary and should still be reviewed against the broader Gaussian-measures literature, where the general distinction between nuclear covariance and radonifying sample operators is classical.

References

- Y. Korolev, J. Latz, C.-B. Schoenlieb, *Gaussian random fields on non-separable Banach spaces*, arXiv:2203.04650.
- N. Weaver, *Lipschitz Algebras*, second edition, World Scientific, 2018. Used here only through Theorem 8.49 as quoted in the source paper.